

Polygraph: Reliable ViT Error Detection from Topological Evidence-Flow

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Abstract

Detecting when an image classifier fails is a fundamental challenge for the safe deployment of Vision Transformers (ViTs). Standard error detection mechanisms often depend primarily on the network’s final output logits, largely ignoring the complex token-to-token routing dynamics that construct the prediction. Drawing inspiration from hallucination detection in large language models, we hypothesize that ViTs exhibit recurring topological signatures when forming incorrect predictions. In this work, we introduce Polygraph, a framework that models a frozen ViT’s attention mechanism as a directed graph to capture these latent failure patterns. By deploying an edge-gated MPNN over the final transformer layer, we extract structural evidence-flow features within a single forward pass. These internal signals are then fused with the full-logit vector via a mixture gate. Evaluated on CIFAR-100-C, the graph model provides complementary predictive signal to strong output- and representation-based baselines, while topology ablations indicate a modest additional benefit from explicit graph structure. The combined Polygraph detector achieves the strongest error-detection performance among methods evaluated in the original benchmark and retains its advantage on corruption families entirely unseen during detector training. In exploratory follow-ups, graph-score fusion improves a fitted LogitDynamics adaptation, while matched set-score fusion attains almost the same point estimate; intermediate auxiliary-head projections provide a smaller improvement within the adaptation.

1 Introduction

Vision Transformers (ViTs) (Dosovitskiy et al., 2021) construct image representations through complex, layer-wise information routing among patch tokens, yet their predictive reliability is commonly estimated from the final softmax distribution. This reliance on output scores leaves much of the

internal token interaction that constructs the prediction unmodeled; predictions with similar output confidence can nevertheless arise from different internal representations and routing patterns.

To address this, we investigate error prediction through the attributed attention graph induced by a ViT. Given a frozen classifier f and an input image x with ground-truth label y , let $\hat{y}(x) = \arg \max_c f_c(x)$, and $e(x) = 1[\hat{y}(x) \neq y]$. Our objective is to predict $e(x)$ using only signals available from a single forward pass of the frozen classifier, without external verification or additional classifier evaluations.

While recent approaches such as LogitDynamics (Beigelman and Freiman, 2026) track the layer-wise trajectory of class evidence, they do not explicitly model the token-to-token connectivity underlying these predictions. We propose Polygraph, a method that represents the final ViT block as a directed attributed graph to capture latent routing signatures. Node features are constructed from token hidden states, while edge features encode evidence-flow statistics derived from attention and class signals. An edge-gated MPNN produces a graph-based internal failure score, which is combined with a strong full-logit detector using a learned per-example mixture gate fitted on held-out validation data. The resulting detector improves over either source of evidence alone and retains its advantage on corruption families excluded from detector training. Controlled topology ablations further show that most of the predictive signal lies in the internal node and edge representations, while graph connectivity and edge alignment provide a more modest additional contribution.

We extend these findings with controlled follow-ups testing competitiveness against a fixed LogitDynamics adaptation, score complementarity, and graph specificity. These studies use distinct development protocols and complement the original non-graph controls.

2 Related Work

ViT Reliability and Attention-Based Diagnostics. A standard baseline for detecting classifier failures is the maximum softmax probability of the predicted class (Hendrycks and Gimpel, 2017). In Vision Transformers, attention and intermediate representations have been exploited for several purposes beyond classification: Chefer et al. (2021) for explanation, Wang et al. (2024) for token pruning, and Yang et al. (2025) for weakly supervised segmentation. Cultrera et al. (2023) use a convolutional autoencoder over ViT attention heatmaps for out-of-distribution detection, but operate on attention heatmaps rather than a token-interaction graph. The closest task-level baseline is LogitDynamics (Beigelman and Freiman, 2026), which predicts ViT classification errors from layer-wise class-logit trajectories and stability statistics derived from intermediate representations, thus reduces each layer to a class-level evidence rather than retaining pairwise interactions among patch tokens.

Graph-Based Error Detection. Natural language processing provides the closest precedents for our work. Earlier approaches construct hand-crafted structural descriptors, such as Kushnareva et al. (2021) and Binkowski et al. (2025). HaluZig (Samaga et al., 2026) models the evolution of attention-graph topology across layers using zigzag persistence. HaluGNN (Kong et al., 2026b) uses hidden states as node features and attention values as weighted edges, while CHARM (Frasca et al., 2026) represents attention and activations as attributes of a token graph and applies edge-aware message passing. CausalGaze (Kong et al., 2026a) further refines attention graphs using gradient-guided counterfactual sensitivity of attention edges. To our knowledge, no prior work applies graph-based message passing to per-image ViT patch-attention graphs for predicting visual classification failures. Polygraph studies this vision-specific setting.

3 Methodology

Polygraph combines two complementary error detectors: one operates on the ViT’s internal attention graph, while the other operates on its output logits. The two experts are trained independently and are subsequently combined by a learned, image-dependent mixture.

3.1 Final-Block Attention Graph

For an image x , let the frozen ViT produce class logits $\ell = f(x) \in \mathbb{R}^C$. We construct a graph from its final Transformer block, containing N tokens and H attention heads. Let $\mathcal{V} = \{0, \dots, N-1\}$ denote the token set, with token 0 representing CLS. For head h , $A_{ij}^{(h)}$ denotes the attention assigned by query token i to key token j . We orient the corresponding edge as $j \rightarrow i$.

We retain an edge whenever it receives sufficiently high attention in at least one head:

$$\mathcal{E} = \left\{ (j, i) : j \neq i, \max_h A_{ij}^{(h)} > \tau \right\} \quad (1)$$

Self-edges are excluded from $\mathcal{E}$, but their diagonal attention values are retained as node attributes.

For each token i , the node feature $\mathbf{x}_i$ concatenates its normalized patch coordinates, a CLS indicator, the diagonal attention values $\{A_{ii}^{(h)}\}_{h=1}^H$, and its final-block hidden representation.

3.2 Evidence-Flow Edge Features

Raw attention describes how strongly token i reads from token j , but not the content or direction of the information being transmitted. We therefore augment every retained edge with two additional per-head features.

Let $\mathbf{v}_j^{(h)}$ denote the value vector of source token j in head h , and let $\mathbf{W}_O^{(h)}$ denote the corresponding part of the multi-head output projection. Their product

$$\mathbf{u}_j^{(h)} = \mathbf{W}_O^{(h)} \mathbf{v}_j^{(h)} \quad (2)$$

represents the contribution of that value after projection back into the Transformer hidden space.

To measure how strongly this contribution supports the model’s current decision, let $\hat{c}$ and $\tilde{c}$ be the classes with the highest and second-highest output logits. If $\mathbf{w}_c$ denotes the frozen classifier weight vector for class c , we define the normalized predicted-versus-runner-up direction

$$\mathbf{d} = \frac{\mathbf{w}_{\hat{c}} - \mathbf{w}_{\tilde{c}}}{\|\mathbf{w}_{\hat{c}} - \mathbf{w}_{\tilde{c}}\|_2} \quad (3)$$

For each retained edge $j \rightarrow i$ and head h , we then compute an attention-weighted magnitude feature and a signed class-direction feature:

$$\begin{aligned} m_{ji}^{(h)} &= \log \left(1 + A_{ij}^{(h)} \left\| \mathbf{u}_j^{(h)} \right\|_2 \right) \\ c_{ji}^{(h)} &= \text{asinh} \left(A_{ij}^{(h)} \mathbf{d}^\top \mathbf{u}_j^{(h)} \right) \end{aligned} \quad (4)$$

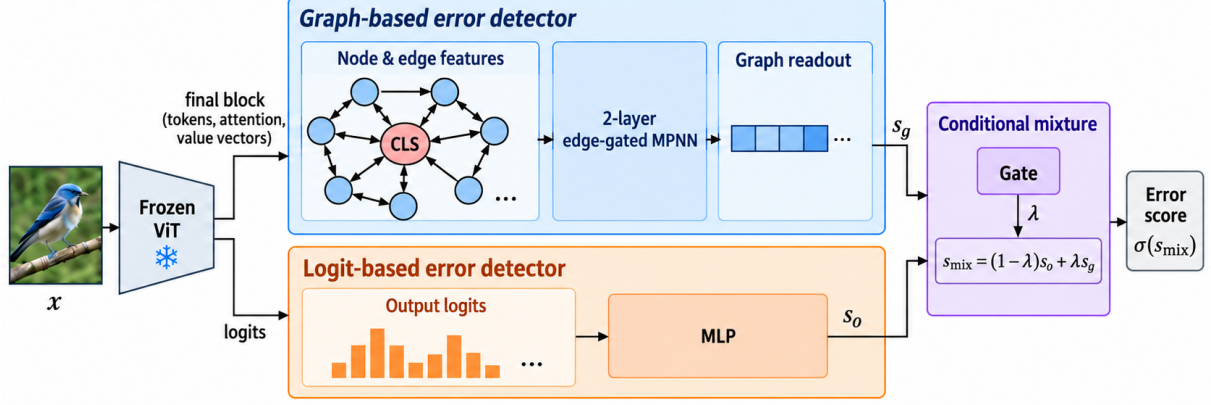


Figure 1: **Overview of Polygraph.** A frozen ViT provides both final-block internal representations and output logits. The graph branch constructs an attributed token graph from the final Transformer block, processes it with a two-layer edge-gated MPNN, and applies graph readout to produce an internal error score s_g . In parallel, a logit-based MLP produces an output-based error score s_o . A conditional mixture gate combines the two experts into the final failure score.

The first captures the magnitude of the transmitted contribution, whereas the second measures whether it aligns with the classifier’s current predicted-versus-runner-up direction. The complete edge attribute $\mathbf{e}_{ji}$ concatenates over all heads the described features. No ground-truth label is used in their construction.

3.3 Edge-Gated Graph Detector

We process the attributed graph using a two-layer edge-gated MPNN. Node and edge attributes are first mapped by learned projections to a shared latent representation.

At message-passing layer r , each edge $j \rightarrow i$ produces both a candidate message and an element-wise gate:

$$\begin{aligned}\mu_{ji}^{(r)} &= M_r([\mathbf{h}_j^{(r-1)} \parallel \tilde{\mathbf{e}}_{ji}]) \\ \gamma_{ji}^{(r)} &= \sigma\left(G_r([\mathbf{h}_j^{(r-1)} \parallel \mathbf{h}_i^{(r-1)} \parallel \tilde{\mathbf{e}}_{ji}])\right)\end{aligned}\quad (5)$$

where $\mathbf{h}_i^{(r)}$ is the latent state of node i after layer r , $\tilde{\mathbf{e}}_{ji}$ is the projected edge attribute and M_r and G_r are learned affine maps.

For the incoming neighborhood $\mathcal{N}_i^- = \{j : (j, i) \in \mathcal{E}\}$, the gated messages are aggregated as

$$\bar{\mu}_i^{(r)} = \frac{\sum_{j \in \mathcal{N}_i^-} \gamma_{ji}^{(r)} \odot \mu_{ji}^{(r)}}{\sum_{j \in \mathcal{N}_i^-} \gamma_{ji}^{(r)}} \quad (6)$$

where $\odot$ denotes element-wise multiplication.

Each node is updated from its previous state, the aggregated message, and its log-in-degree

$\log(1 + |\mathcal{N}_i^-|)$, using an MLP with dropout, a residual connection, LayerNorm, and ReLU.

After message passing, we combine the final CLS-node state with a learned weighted pooling of all token states. Let ψ be a learned scalar node scorer and ρ the final graph decoder. The graph error logit is

$$\begin{aligned}\beta_i &= \frac{\exp(\psi(\mathbf{h}_i))}{\sum_{j \in \mathcal{V}} \exp(\psi(\mathbf{h}_j))} \\ s_g(x) &= \rho\left([\mathbf{h}_0 \parallel \sum_{i \in \mathcal{V}} \beta_i \mathbf{h}_i]\right)\end{aligned}\quad (7)$$

Thus, the detector retains direct access to the CLS token while also learning which patch-token representations are informative for failure prediction.

3.4 Output Expert and Conditional Fusion

The second expert operates directly on the complete output-logit vector and produces an error logit $s_o(x)$. The output MLP receives the full logit vector, standardized per class using detector-training statistics. The graph and output experts are trained independently using class-weighted binary cross-entropy against

$$Y_{\text{err}}(x) = 1[\hat{c} \neq y] \quad (8)$$

where y is the ground-truth class.

Once trained, both experts are frozen. Their scores are independently calibrated on held-out meta-validation data, producing calibrated logits

$z_o(x)$ and $z_g(x)$. A small gate receives the standardized pair of expert scores and predicts an example-specific graph weight $\lambda(x) \in [0, 1]$:

$$z_{\text{mix}}(x) = (1 - \lambda(x)) z_o(x) + \lambda(x) z_g(x) \quad (9)$$

Equation (9) is applied separately in each of five grouped meta-validation folds. Each fold fits its score standardizer, scalar logistic calibrators, and gate on four folds, using the remaining fold for gate early stopping. At test time, we average the five mixture logits; both frozen experts are shared across folds.

The gate therefore learns when the internal graph signal provides useful information beyond the output-logit expert. To avoid leakage, calibration and gate fitting are performed only on a disjoint meta-validation split using grouped cross-validation by original image.

Implementation details. In our experiments, the ViT has $N = 197$ tokens, $H = 12$ heads, $D = 768$, and per-head dimension 64. We use $\tau = 0.02$, resulting in approximately 7,600 directed edges per image. Node features contain 16 base attributes plus the 768-dimensional hidden state, and edge attributes contain 36 values (12 attention, 12 magnitude, and 12 class-direction features). The graph detector uses latent width 64, two message-passing layers, and dropout 0.15. The output expert has dimensions $100 \rightarrow 64 \rightarrow 32 \rightarrow 1$, and the mixture gate uses $2 \rightarrow 16 \rightarrow 1$. We use five grouped folds for mixture fitting.

4 Experimental Setup

This section describes the original Polygraph experiments; the follow-ups use the distinct protocols in Section 6.

4.1 Datasets and Splits

We evaluate on CIFAR-100 and CIFAR-100-C using a frozen ViT-B/16 pretrained on ImageNet-21k and fine-tuned on CIFAR-100. We construct an error-balanced benchmark by sampling equal numbers of correct and incorrect predictions within each source-severity stratum. All versions of the same original image remain in one train/validation/test partition. We consider the main benchmark and a weather-family holdout in which selected corruption families are excluded from detector training and used to evaluate distribution-shift generalization. Model selection uses only the

corresponding validation split, and results for these original experiments are on held-out test data.

4.2 Training and Baselines

We select models by validation AUROC with early stopping and repeat all learned detectors over three random seeds. We compare against maximum softmax probability (MSP) and a strict output-only MLP trained on the complete classifier-logit vector.

5 Results

5.1 Main Error Detection

Table 1 compares detectors on the main CIFAR-100/CIFAR-100-C test split. Polygraph improves mean AUROC over the strict full-logit MLP, while also improving AUPRC and AURC. The standalone graph expert provides a smaller mean AUROC gain. Throughout this section, *graph expert* denotes the selected two-layer edge-gated MPNN, and *Polygraph* denotes its conditional fusion with the output MLP.

Method	AUROC $\uparrow$	AUPRC $\uparrow$	AURC $\downarrow$
MSP	0.8651	0.8345	0.2218
Output MLP	0.8794	0.8583	0.2158
Graph expert	0.8883	0.8682	0.2105
Polygraph	0.8939	0.8748	0.2071

Table 1: Main-test performance, averaged over three seeds for learned detectors. MSP is deterministic. Bold marks the best result in each column.

Using 2,000 paired bootstrap repetitions grouped by original image, the 95% confidence interval for the Polygraph-output AUROC difference lies above zero in all three seeds. The standalone graph improvement is less certain, although at an equal flagging budget of 50% of test examples per detector, the graph expert identifies 37.4% of the true errors missed by the output MLP, averaged across seeds. These distinct error sets support using the graph score as complementary evidence.

Corruption severity. Table 2 reports the main-test breakdown by corruption severity. Polygraph has the highest mean AUROC at each severity from 1 to 5. The graph expert alone also exceeds the output MLP at every reported severity.

5.2 Held-Out Corruption Generalization

Under the weather-family holdout, Polygraph improves over the output MLP on both the com-

Severity	MSP	Output MLP	Graph expert	Polygraph
1	0.8928	0.8967	0.8978	0.9065
2	0.8904	0.8977	0.9047	0.9115
3	0.8727	0.8804	0.8905	0.8957
4	0.8537	0.8743	0.8857	0.8907
5	0.8184	0.8537	0.8673	0.8707

Table 2: Main-test AUROC by corruption severity, pooling corruption types within each severity. Learned-detector values are three-seed means.

plete test set and the subset containing only unseen weather and extra corruptions (Table 3). On the unseen subset, Polygraph also exceeds MSP, whereas the standalone graph expert and the output MLP do not exceed MSP. Thus, conditional fusion retains its benefit for corruption types excluded from detector training.

Method	Full test AUROC $\uparrow$	Unseen only AUROC $\uparrow$
MSP	0.8625	0.8844
Output MLP	0.8690	0.8816
Graph expert	0.8764	0.8840
Polygraph	0.8820	0.8912

Table 3: Weather-holdout AUROC, averaged over three seeds for learned detectors. Full test includes seen and unseen corruption families; Unseen only contains the held-out weather and extra corruptions.

5.3 Representation and Topology Ablations

Feature ablations. The TransformerConv feature study (Table 4) separates the effects of node and edge attributes. With base node features, adding class direction raises AUROC substantially, whereas message magnitude alone adds little. Combining magnitude with direction does not improve over direction alone. With full hidden-node inputs held fixed, replacing raw-attention edges with full evidence-flow features additionally increases AUROC.

Non-graph controls. Table 5 compares models under disjoint validation data for expert selection and fusion. All models receive the same hidden-node inputs. The node/edge-set control pools node and edge features separately, without retaining their incidence; the endpoint-set control retains local source–target–edge records but performs no message passing. Both nearly match TransformerConv. The edge-gated MPNN has the highest mean AU-

Nodes	Edge features	AUROC $\uparrow$
Base	Attention	0.8348
Base	Attention + magnitude	0.8369
Base	Attention + direction	0.8660
Base	Full evidence flow	0.8655
Compact	Full evidence flow	0.8773
Hidden	Attention	0.8754
Hidden	Full evidence flow	0.8863

Table 4: Feature ablations with TransformerConv (seed 7, original validation split). Compact nodes add class-conditioned summaries; hidden nodes add full token states. Full evidence flow includes attention, magnitude, and class direction.

ROC.

Internal detector	AUROC $\uparrow$
TransformerConv	0.8845
Token set, no edges [†]	0.8838
Node/edge sets, no incidence	0.8834
Endpoint sets, no message passing	0.8813
Edge-gated MPNN	0.8883

Table 5: Main-test topology controls under the strict validation protocol. All models use hidden-node inputs. Values are three-seed means except [†], which uses seed 7 only.

5.4 Architecture and Depth

Validation screening favored the edge-gated MPNN over the tested TransformerConv, GINE, and GATv2 variants. Their validation AUROCs were 0.8849, 0.8763, 0.8753, and 0.8759, respectively. The selected edge-gated model has 132k parameters, compared with 191k for the TransformerConv model. For the edge-gated architecture, two message-passing layers achieved 0.8849 validation AUROC, compared with 0.8792 for one layer and 0.8789 for four.

Extending the representation to the final four ViT blocks did not improve the tested models. 4-layer variants achieved test AUROCs of 0.8771 and 0.8820, respectively, versus 0.8894 for the final-block edge-gated expert at the same seed. These results support the selected configuration within the evaluated architectures.

6 Controlled Follow-up Studies

We retain the frozen ViT-B and group each source photograph’s nine views: clean, plus Gaussian noise, motion blur, fog, and JPEG at severities 3/5.

Unlike the original benchmark, these cohorts are not error-balanced. Subsequent comparisons reuse examined development data; they are exploratory, not independent confirmation.

When does graph processing help? A separate five-seed study found a rich36 graph-set AUROC difference of $+0.006233$ (95% interval $[0.002618, 0.009836]$), versus -0.003686 ($[-0.008349, 0.001045]$) with attention-only inputs. Rich36 combines per-head attention, projected-message magnitude, and signed predicted-versus-runner-up class support. A later raw12 study instead trains separate detectors on attention layers 3/6/9/12, all sharing H12 token states. Both arms receive identical 784-dimensional node values and 12 attention edge values; retained edges zero-fill head values at or below 0.02, unlike the original all-head representation. The graph arm uses two width-64 edge-gated blocks with CLS/gated pooling; the width-84 set arm pools node/edge values separately, discarding endpoint associations. Parameters are closely matched (130,434/131,126), but endpoint treatment and pooling differ.

The later detectors complete 20 epochs on 1,600 training photographs, selecting by AUROC on 400 others. Their within-seed mean sigmoid scores define G and S ; prior graph fits are reused. Their AUROC gap is $+0.000336$ (six-contrast-adjusted 99 $\frac{1}{6}$ % interval $[-0.002800, 0.003409]$); G reaches 0.888895, versus 0.883127 for the fitted single-final-layer control. Separate detectors followed by score combination differ from the joint multi-block representations above. No matched four-model final-layer ensemble isolates layer diversity. Cohorts, features, budgets and constructions differ across studies, precluding a controlled replication comparison or attribution of the changed gap to richness or ensembling.

LD comparison and readout decomposition. Our fixed LogitDynamics (LD) adaptation (Beigelman and Freiman, 2026) trains linear class heads on all 12 CLS layers, then a standardized linear error readout. This is a ViT-Base adaptation, not a reproduction of the published ViT-Large study or the earlier local CLS-sequence GRU baseline. We allocate 1,200/800/400/800 source photographs to head fitting/readout fitting/checkpoint selection/development evaluation. Heads complete 16 epochs; readouts complete 100 with validation average-precision (AP) selection. Seeds are

7/17/27. The shared evaluation has 7,200 views and 2,271 errors; families differ in inputs, supervision, parameters and budgets.

Model	AUROC
<i>Preserved comparator families</i>	
Graph-score ensemble G	0.8889
Set-score ensemble S	0.8886
Full-logit MLP O	0.8757
<i>Nested LD readouts</i>	
A : six original-classifier features	0.8636
B : A + layer-12 auxiliary block	0.8948
C : all 12 auxiliary blocks + original	0.8990
D : C + seven dynamics features	0.8981

Table 6: Mean seed-specific AUROC on the same 800 development photographs. Comparator families are not matched to the nested LD ablation. Only readouts $A/B/C$ are newly fitted; auxiliary heads and full D are preserved.

Each numeric block contains the original classifier’s predicted-class logit and five largest competing logits. A is therefore not full-logit MLP O . Only $A/B/C$ are newly fitted, reusing heads and full adaptation D (Table 6). The earlier $G - D$ contrast is -0.009171 (ordinary 95% interval $[-0.015874, -0.002361]$), favoring LD over this preserved ensemble, not comparing LD to original rich conditional Polygraph. $B - A = +0.031233$ is the largest observed ablation increment. Intermediate projections add $C - B = +0.004135$ (adjusted $[0.000428, 0.007911]$), with a lower bound close to zero. $D - C = -0.000900$ (secondary 95% $[-0.001750, -0.000048]$): the dynamics block, including class-identity information, did not improve this fitted recipe. These increments neither isolate supervision causally nor explain the entire LD-GNN gap.

Do weaker scores still complement LD? We split the examined 800 photographs into 400 for combiner fitting and 400 for assessment, preserving all views. For each seed, identical standardized L2 logistic regressions combine $D + G$, $D + S$, or $D + D'$. Scaling/fitting use only the fitting subset. D is the preserved pre-sigmoid error logit, left uncalibrated; G/S are mean probabilities. This is not the original calibrated nonlinear five-fold mixture gate. Cyclic D' pairs (7–17, 17–27, 27–7) overlap and are not independent repetitions.

Graph fusion improves LD by $+0.007203$ (adjusted $[0.003415, 0.011330]$), despite weaker standalone G . However, $DG - DS = +0.000043$ (adjusted $[-0.001311, 0.001508]$) shows no demon-

Model	AUROC
LD D	0.9001
Graph ensemble G	0.8956
Set ensemble S	0.8950
$D + G$	0.9073
$D + S$	0.9073
$D + D'$	0.9021

Table 7: Original fitted models’ mean seed-specific AUROC on the separate 400-photo fusion assessment (3,600 views). Values are not directly comparable with Table 6 or the original benchmark.

strated graph-specific benefit (Table 7). In the secondary comparison, graph-LD fusion also outperformed the tested two-seed LD combination: $DG - DD' = +0.005213$ (95% [0.001942, 0.008673]). This control is not compute-matched and establishes neither superiority over generic ensembling generally nor a graph-specific benefit.

Uncertainty and scope. AUROC treats error as positive and averages seed-specific metrics, not merged predictions. Resampling pairs source photographs, retaining nine views and shared draws across methods/seeds. The two new studies use 2,000 draws: fusion independently resamples fitting/assessment photographs and refits combiners; decomposition resamples evaluation only. Intervals condition on fixed base detectors/partition or all fitted readouts, respectively. Bonferroni adjustment across $DG - D$, $DG - DS$, and $C - B$ yields individual $98\frac{1}{3}\%$ percentile intervals; secondary 95% intervals are descriptive. Neither correction nor additional seeds remove development-data exposure. Nonsignificance is not equivalence, and complementarity does not establish topology’s necessity.

7 Conclusion

We introduced Polygraph, a graph-based framework for error detection in Vision Transformers that models token interactions as an evidence-flow graph and combines this internal signal with the classifier output through a conditional gate. The resulting model improves over the strongest output-only baseline and remains competitive under distribution shift. Exploratory follow-ups show graph/set scores complementing LD without a demonstrated graph-specific fusion advantage. Together, these results suggest that the tested internal-feature scores provide complementary information to output confidence and can improve failure prediction in vision transformers.

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